

Chapter: 1

sets:

unique

Defⁿ: a collection of things $\rightarrow$ anything

represent: $\{a, b, c, d\}$ $\rightarrow$ elements

no of elements = 4

Defⁿ: a set whose no of elements is finite
finite set.

example: $\{1, 2, 5, 7, 8, 10\}$

Defⁿ: a set whose no of element is 'infinite'

$$\left\{ \begin{array}{l} \mathbb{N} = \{1, 2, 3, 4, 5, 6, 7, \dots\} \\ \mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\} \\ \mathbb{Q} = \{p/q, p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0, \boxed{p \text{ \& } q \text{ co primes}}\} \end{array} \right.$$

$$\mathbb{Q}^c =$$

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$$

lets consider two sets A & B

$$A = \{1, 2\}, \quad B = \{2, 1\}$$

From her point of view they are basically the same thing,

$$\{2, 4, 6, 8\} = \{4, 6, 2, 8\}$$

$$\{2, 4, 6, 8\} \xrightarrow{\text{element}} A \quad 2 \in A \quad 5 \notin A$$

Probable outcomes of a coin toss,

$$C = \{H, T\}$$

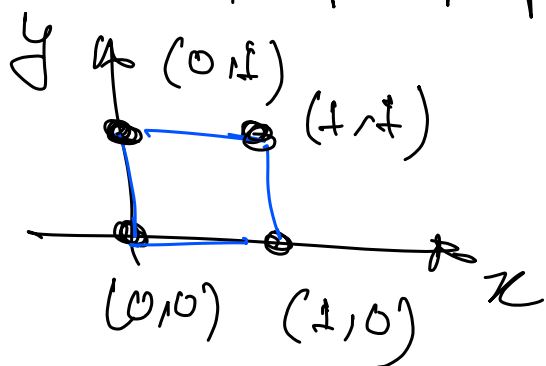
Probable outcomes of a dice

$$D = \left\{ \begin{array}{|c|} \hline \cdot \\ \hline \end{array}, \begin{array}{|c|} \hline \cdot \cdot \\ \hline \end{array}, \begin{array}{|c|} \hline \cdot \cdot \cdot \\ \hline \end{array}, \begin{array}{|c|} \hline \cdot \cdot \cdot \cdot \\ \hline \end{array}, \begin{array}{|c|} \hline \cdot \cdot \cdot \cdot \cdot \\ \hline \end{array}, \begin{array}{|c|} \hline \cdot \cdot \cdot \cdot \cdot \cdot \\ \hline \end{array} \right\}$$

Collection of vowels,

$$E = \{a, e, i, o, u\}$$

Collection of 4 points of a square in $\mathbb{R}^2$



$$D = \{(0,0), (0,1), (1,1), (1,0)\}$$

Set consisting of sets as elements

$$E = \{ \{1, \{1, 2\}\}, 2, \{2\} \}$$

ordered pairs:

$$E = \{ (a, b), 2, e, \{1\} \}$$

a counter example:

$$A = \{ 2, 3, 4, 5 \} \quad 2 \in A$$

$$\bullet E = \{ \{1, \{1\}\}, \{1, 2\}, \{2\} \}$$

$$\bullet 1 \in E? \quad \bullet 2 \in E?$$

Cardinality:

No of elements of a set,

$$X = \{ a, b, c, e, g \}$$

$$|X| = 5$$

$$C = \{ H, T \}, \quad |C| = 2$$

• What does it mean for a set to have 0 cardinality?

construction: $\{ \}$ empty sets: $\emptyset, \{ \}$

Common misconceptions:

$$|\emptyset| = 0$$

$$|\{\emptyset\}| = 1$$

example:

$$F = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}\}$$

$$|F| = 3$$

another example:

$$P = \{\mathbb{N}, \mathbb{Z}\}$$

$$|P| = 2$$

natural integer

Roaster method

$$\{a, b, c, d, e, \dots\}$$

Drawback:

(i) lengthy

(ii) It is sometimes confusing

$$\{3, 5, 7, \dots\}$$

Set builder method:

$$E = \{\text{expression: rule}\}$$

$$E = \{-4, -2, 0, 2, 4, 6, 8, \dots\}$$

$$E = \{2n : n \in \mathbb{Z}\}$$

• in Roaster method you can express a set multiple way:

$$A = \{1, 2\} = \{2, 1\}$$

$$\mathbb{Z} = \{-4, -3, -2, -1, 0, 1, 2, 3, 4, \dots\}$$

$$\mathbb{Z} = \{0, 1, -1, 2, -2, 3, -3, \dots\}$$

similarly in set builder method:

$$E = \{2n : n \in \mathbb{Z}\}$$

$$E = \{n : n = 2k, k \in \mathbb{Z}\}$$

example :

$$\{n : n \text{ is a prime}\} =$$

$$4. \{x \in \mathbb{Z} : x - 2 = 0\} = \emptyset$$

$$\{x \in \mathbb{N} : -2 \leq x \leq 7\} = \{-1, 2, 3, 4, 5, 6, 7\}$$

$$\{ \underline{2x} : x \in \mathbb{Z}, |x| < 4 \} =$$

$$\{ -3, -2, -1, 0, 1, 2, 3 \}$$

$$\{ x \in \mathbb{Z}, |2x| < 4 \} = \{ -1, 0, 1 \}$$

$$\bullet \textcircled{A} = \{ \underline{6a + 2b}, a \in \mathbb{Z}, b \in \mathbb{Z} \}$$

a, b : integers

$6a$: even $2b$: even

$6a + 2b$: even

A is a set which consists of even integers

$$A = \underline{2n} \quad 2n = \underline{6 \cdot n + 2 \cdot (-2n)}$$

$$\underline{a = n}, \quad b = -2n$$

Hence, A consists of all even integers

infinite set:

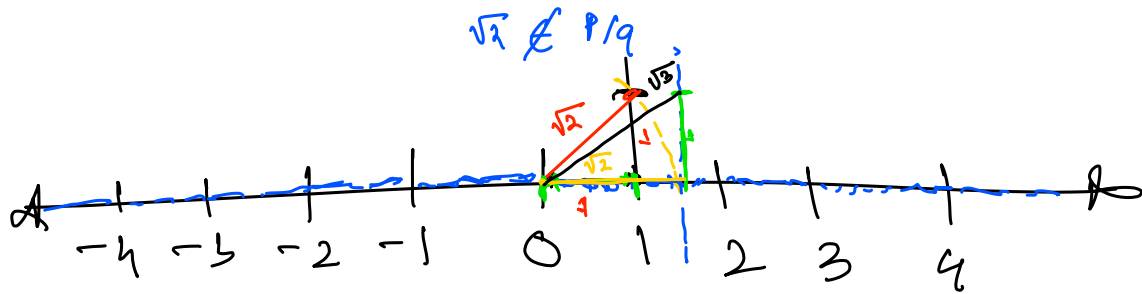
$$(N = \{ 1, 2, 3, 4, 5, 6, \dots \})$$

$(+, \cdot)$

$$\mathbb{Z} = \{ \dots -3, -2, -1, 0, 1, 2, 3, 4, \dots \}$$

$$\mathbb{Q} = \left\{ \frac{p}{q}, p \in \mathbb{Z}, q \in \mathbb{Z}, q \neq 0, \boxed{p \text{ \& not\& } q \text{ coprime}} \right\}$$

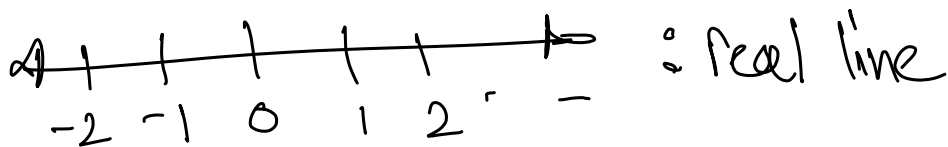
$$\frac{2}{3} = \frac{4}{6} = \frac{8}{12} = \dots$$



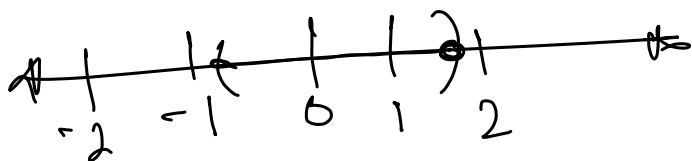
$$\mathbb{Q}^c = \left\{ \sqrt{2}, \sqrt{3}, \sqrt{6}, \pi, \sqrt[3]{7}, \dots, -\sqrt{2}, \gamma, \frac{1}{\gamma}, \frac{1}{\pi}, \pi^{\sqrt{2}} \right\}$$

• Completion

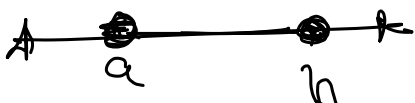
$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^c$$

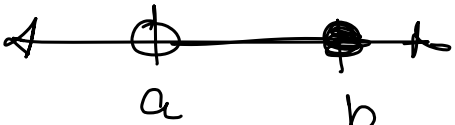


Regions in real lines:

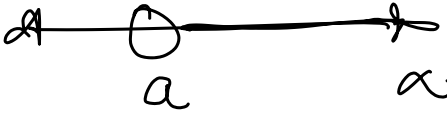


open interval  (a, b)

closed interval  $[a, b]$

 $(a, b]$

 $[a, b)$

 (a, ∞)

 $[a, \infty)$

 $(-\infty, b)$

 $(-\infty, b]$

exercice:

37.

$$|\{x \in \mathbb{N} : x < 0\}| = |\emptyset| = 0$$

$$|\{ \{1, 4\}, a, b, \{3, 4\} \}| = 4$$

Cartesian Product :

Order pair : (a, b)

• What is the difference between order pair & set?

→ Sequence matters

$$(a, b) \neq (b, a)$$

→ (a, a) $\{a, a\}$

elements of order pair

$(2, \{2, 3\})$, $(2, m)$,

$(\mathbb{R}, (0, 0))$